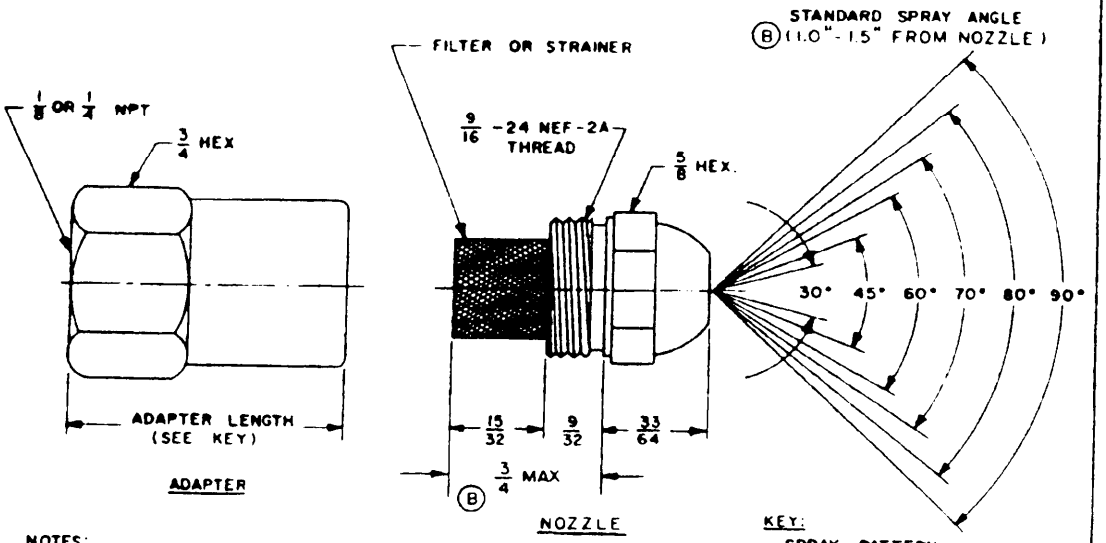
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REVIEW ACTIVITIES: AF - 99
USER ACTIVITIES: ARMY-CE



KEY:
SPRAY PATTERN
S = SOLID
SH = SEMI-HOLLOW
H = HOLLOW
ADAPTER LENGTH
S = SHORT (15/16)
L = LONG (1-3/8)
X = EXTRA LONG (2 - 2 1/16) (B)

NOTES:

1. MATERIALS
   (B) ADAPTER - BRASS
   NOZZLE - CORROSION RESISTANT STEEL FUNCTIONAL PARTS
   FILTER - BRONZE
   SCREEN STRAINERS - NICKEL COPPER ALLOY
2. THE PART NUMBER CONSISTS OF THE MS NUMBER AND A DASH NUMBER TO INDICATE THE (a) CAPACITY, (b) SPRAY PATTERN, (c) ANGLE OF SPRAY, (d) ADAPTER WITH THREAD SIZE, AND (e) ADAPTER LENGTH.
3. THE NOZZLE SHALL BE FURNISHED WITH AN ADAPTER WHEN "A" IS ADDED FOLLOWING THE ANGLE OF SPRAY. SPECIFY PIPE THREAD, 1/8 OR 1/4 NPT REQUIRED ON THE NOZZLE ADAPTER BY INCLUDING A "8" FOR 1/8 NPT OR A "4" FOR 1/4 NPT WITH THE "A" FOR ADAPTER. SPECIFY LENGTH OF ADAPTER BY "S", "L" OR "X" PER KEY ABOVE

   EXAMPLE: MS 49007 - 02 H 30 A8 L
   (a) (b) (c) (d) (e)

   (a) 02 = 0.50 RATED CAPACITY IN GPH @ 100 PSI; INDICATED IN COL. (a) TABLE I.
   (b) H = HOLLOW SPRAY PATTERN; INDICATED IN KEY.
   (c) 30 = 30° ANGLE OF SPRAY; INDICATED IN TABLE II.
   (d) A8 = ADAPTER WITH 1/8 NPT WILL BE FURNISHED WITH THE NOZZLE.
   (e) L = LONG ADAPTER (1 3/8).
4. DIMENSIONS ARE IN INCHES.
   TOLERANCES
   FRACTIONAL $\pm \frac{1}{32}$
   ANGULAR $\pm 5°$
5. (B) THREAD SIZES SHALL CONFORM TO FED-STD-H28.
6. THE SEALING SURFACES BETWEEN THE ADAPTER AND THE NOZZLE SHALL HAVE A 16 MICROINCH FINISH OR BETTER.

(B) INDICATES CHANGES

APPROVED 11 JANUARY 1963 REVISED (A) 6 MARCH 1967 (B) 30 MARCH 1992

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. ARMY ME Other Cust AF - 69 NAVY - YD	TITLE NOZZLES, OIL BURNER PRESSURE ATOMIZING	MILITARY STANDARD MS 49007
PROCUREMENT SPECIFICATION NONE	SUPERSEDES	SHEET 1 OF 3

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4530-0003

AMSC N/A

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TABLE I

U.S. GALLONS PER HOUR NO. 2 FUEL OIL *

(a)	RATED GPH @ 100 PSI	OPERATING PRESSURE PSI				
		75	125	150	175	200
1	0.40	(B)	0.45	0.49	0.53	0.56
2	0.50		0.56	0.61	0.66	0.71
3	0.60		0.67	0.74	0.79	0.85
4	0.65		0.73	0.80	0.86	0.92
5	0.75		0.84	0.92	0.99	1.06
6	0.85		0.95	1.04	1.13	1.20
7	1.00		1.12	1.23	1.32	1.41
8	1.10		1.23	1.34	1.45	1.55
9	1.20		1.34	1.47	1.59	1.70
10	1.25		1.39	1.53	1.65	1.77
11	1.35		1.51	1.65	1.79	1.91
12	1.50		1.68	1.84	1.98	2.12
13	1.65		1.84	2.02	2.18	2.34
14	1.75		1.96	2.14	2.32	2.48
15	2.00		2.24	2.45	2.65	2.83
16	2.25		2.52	2.74	2.98	3.18
17	2.50		2.80	3.06	3.30	3.54
18	3.00		3.35	3.68	3.97	4.25
19	3.50		3.91	4.29	4.63	4.95
20	4.00		4.47	4.90	5.30	5.66
21	4.50		5.04	5.51	5.95	6.36
22	5.00		5.59	6.13	6.61	7.07
23	5.50		6.15	6.74	7.27	7.78
24	6.00		6.71	7.33	7.94	8.48
25	6.50		7.26	7.96	8.60	9.20
26	7.00		7.82	8.58	9.25	9.90
27	7.50		8.38	9.19	9.91	10.60
28	8.00		8.94	9.79	10.58	11.31
29	8.30		9.28	10.20	11.00	11.80

(a)	RATED GPH @ 100 PSI	OPERATING PRESSURE PSI				
		75	125	150	175	200
30	9.00	(B)	10.06	11.02	11.91	12.73
31	9.50		10.60	11.70	12.60	13.50
32	10.00		11.18	12.25	13.23	14.14
33	10.50		11.70	12.90	13.90	14.90
34	11.00		12.29	13.47	14.55	15.55
35	12.00		13.40	14.70	15.90	17.00
36	13.80		15.40	16.90	18.30	19.60
37	14.00		15.65	17.15	18.52	19.79
38	15.30		17.10	18.70	20.30	21.60
39	16.00		17.89	19.60	21.17	22.63
40	17.50		19.60	21.40	23.20	24.80
41	18.00		20.12	22.04	23.81	25.48
42	19.50		21.80	23.90	25.80	27.60
43	20.00		22.36	24.49	26.46	28.28
44	21.50		24.00	26.40	28.40	30.40
45	22.00		24.60	26.94	29.10	31.11
46	24.00		26.80	29.40	31.80	34.00
47	26.00		29.07	31.84	34.39	36.77
48	28.00		31.30	34.30	37.00	39.60
49	30.00		33.60	36.80	39.70	42.50
50	35.00		39.10	42.90	46.30	49.50
51	40.00		44.70	49.00	53.00	56.50
52	45.00		50.40	55.20	59.50	63.70
53	50.00		55.90	61.30	66.10	70.70
54	55.00		61.50	67.40	72.70	77.70
55	60.00		67.00	73.50	79.40	84.00
56	70.00		78.20	85.70	92.50	99.00
57	80.00		89.40	98.00	106.00	113.50
58	90.00		100.90	110.50	119.20	127.50
59	100.00		111.90	122.50	132.30	141.40

* BASED ON OIL WEIGHING 7.16 LBS PER GALLON @ 74°F AND 34 SSU VISCOSITY AT 100°F

APPROVED 11 JANUARY 1963 REVISED (B) 30 MARCH 1992

FOR CHANGES SEE SHEET 1

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TABLE II

ANGLE OF SPRAY SPRAY PATTERN- CAPACITY RANGE			
ANGLE OF SPRAY	SPRAY PATTERN (B)		RATED CAPACITY GPH @ 100 PSI
30°	SOLID	(S)	.50 TO 24.00
	SEMI-SOLID	(SH)	4.50 TO 20.00
	HOLLOW	(H)	.50 TO 8.00
45°	SOLID	(S)	.40 TO 30.00
	SEMI-SOLID	(SH)	2.25 TO 50.00
	HOLLOW	(H)	.50 TO 50.00
60°	SOLID	(S)	.40 TO 4.00
	SEMI-SOLID	(SH)	2.25 TO 50.00
	HOLLOW	(H)	.40 TO 50.00
70°	SOLID	(S)	.50 TO 60.00
	SEMI-SOLID	(SH)	2.25 TO 60.00
	HOLLOW	(H)	.50 TO 35.00
80°	SOLID	(S)	.40 TO 100.00
	SEMI-SOLID	(SH)	2.25 TO 100.00
	HOLLOW	(H)	.40 TO 50.00
90°	SOLID	(S)	.60 TO 50.00
	SEMI-SOLID	(SH)	2.25 TO 50.00
	HOLLOW	(H)	.60 TO 9.50

TABLE III

FILTERS & STRAINERS	
RATED CAPACITY	TO BE FURNISHED WITH NOZZLE
.40 TO 1.00	POROUS BRONZE FILTER (25 TO 50 MICRONS)
.60 TO .85	SCREEN STRAINER 200 MESH
1.10 TO 2.00	POROUS BRONZE FILTER (75 TO 100 MICRONS)
1.10 TO 10.58	SCREEN STRAINER 100 TO 120 MESH
11.00 TO 100.00	NO FILTER

NOTES:

(B) THE SCREEN STRAINER CAN BE SUBSTITUTED FOR THE POROUS BRONZE ON THE FOLLOWING FLOW RATES:

.40 - 1.00 AND 1.10 - 2.00

APPROVED 11 JANUARY 1961 REVISED (A) FOR CHANGES SEE SHEET 1 (B) 30 MARCH 1992

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P.A. ARMY - ME	TITLE	MILITARY STANDARD
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